

PROJECT DOCUMENTATION

Supply Chain Analytics Dashboard

End-to-End Supply Chain Analytics using Microsoft Power BI — from raw operational data to executive-ready, decision-grade dashboards.

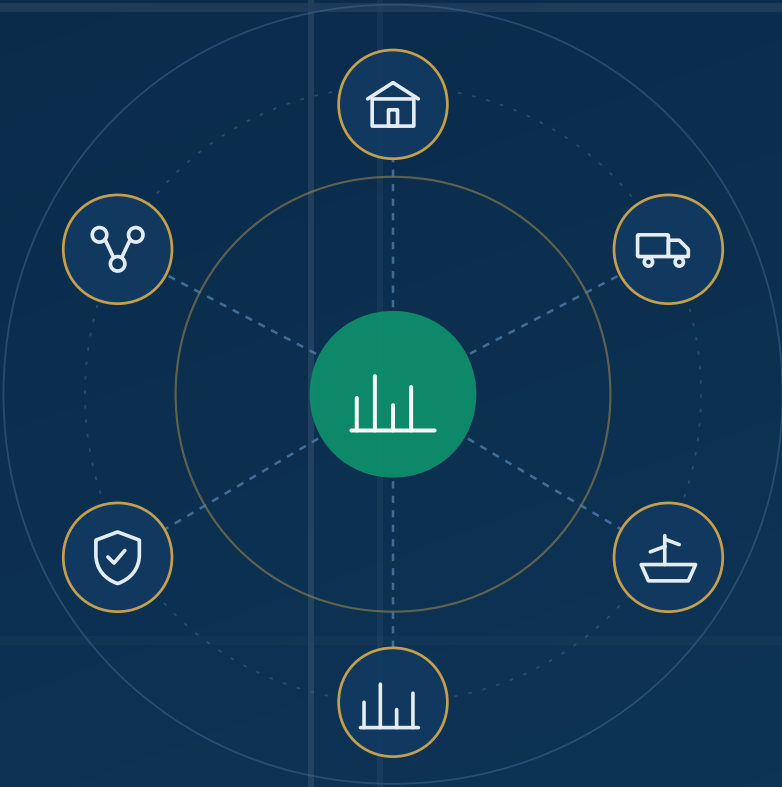


Table of Contents

Supply Chain Analytics Dashboard — Power BI Project Documentation

01	Executive Summary	4
02	Project Overview	5
03	Dataset Description	7
04	Project Workflow	10
05	Data Cleaning Process	11
06	Data Modeling	13
07	DAX Measures	15
08	Dashboard Development	19
09	Visualization Techniques	27
10	Business Insights	28
11	Business Recommendations	29
12	Project Outcomes	30
13	Challenges Faced	32
14	Future Enhancements	33
15	Skills Demonstrated	34
16	Conclusion	35
17	References	36
18	GitHub Project Structure	37
A	Appendix A: Sample Data Preview	38
B	Appendix B: KPI Glossary	39

Page references correspond to the final compiled document and may vary slightly by viewer/renderer.



Project Foundations

Executive summary, project overview, dataset description, workflow, data cleaning, data modeling, and the DAX measure library that underpins every dashboard in this report.

01 Executive Summary



A concise overview of the Supply Chain Analytics Dashboard project, its objectives, the business problem it addresses, and the value it delivers to decision-makers.

Project Overview

The **Supply Chain Analytics Dashboard** is an end-to-end Business Intelligence solution built on **Microsoft Power BI** that transforms raw, transaction-level supply chain data into a suite of interactive, decision-ready dashboards. The project ingests a real-world style supply chain dataset covering product, supplier, manufacturing, shipping, inventory, and quality-inspection records, and converts it into governed data models, calculated business measures, and visual analytics that support operational, tactical, and strategic decision-making across the supply chain function.

The solution was designed to mirror how a mid-sized consumer goods organization (spanning the *haircare*, *skincare*, and *cosmetics* product categories) would monitor revenue performance, supplier reliability, logistics efficiency, product quality, and customer demand in a single, governed reporting environment — replacing fragmented spreadsheet-based reporting with a single source of truth.

Objectives

- Consolidate fragmented supply chain data (sales, manufacturing, logistics, quality, and supplier records) into a single, well-modeled analytical dataset.
- Design a scalable star-schema data model to support fast, flexible analysis in Power BI.
- Develop a comprehensive DAX measure library covering revenue, cost, inventory, logistics, and quality KPIs.
- Build eight purpose-built dashboards addressing executive, sales, inventory, logistics, quality, supplier, forecasting, and customer analytics use cases.
- Translate raw data into actionable business insights and evidence-based recommendations.
- Demonstrate industry-standard BI development practices suitable for professional and academic portfolio presentation.

Business Problem

Supply chain organizations generate large volumes of operational data across disconnected systems — sales ledgers, manufacturing logs, supplier scorecards, shipping records, and quality inspection reports. Without a unified analytics layer, decision-makers are forced to rely on static, manually assembled spreadsheets that are slow to refresh, prone to error, and incapable of surfacing cross-functional insight (for example, the relationship between supplier lead time and stock-out risk, or between transportation mode and shipping cost). This project addresses that gap by building a governed, automated, and visually intuitive analytics layer directly on top of the raw operational dataset.

Expected Business Value

\$577.6K

TOTAL REVENUE ANALYZED

46,099

UNITS SOLD (DATASET)

100

SKUS MODELED

8

DASHBOARDS DELIVERED

- ✓ Faster, self-service decision-making — stakeholders explore data directly instead of waiting on manual reports.
- ✓ Improved visibility into supplier reliability, product quality, and logistics cost drivers.
- ✓ A single, trusted source of truth for revenue, inventory, and operational KPIs across the organization.
- ✓ A reusable, extensible analytics framework that can absorb new data sources (e.g., real-time ERP feeds) with minimal rework.
- ✓ Reduced manual reporting effort, freeing analysts to focus on interpretation and strategy rather than data assembly.

🔍 Why This Project Matters

This project was designed to demonstrate the complete Business Intelligence lifecycle — from raw data acquisition through cleaning, modeling, DAX measure design, dashboard development, and business storytelling — using the same standards expected in a professional analytics team. It is intended as a portfolio-grade artifact suitable for GitHub publication, technical interviews, and stakeholder presentations.

02 Project Overview



Introduction

Modern supply chains generate a continuous stream of operational data spanning procurement, manufacturing, logistics, and sales. Turning this data into a coherent, actionable narrative requires more than spreadsheets — it requires a structured BI development process: reliable data ingestion, careful cleaning and transformation, a well-designed data model, a governed set of business calculations (DAX measures), and a visualization layer that is tailored to different organizational audiences. This project implements that full process using Microsoft Power BI as the delivery platform.

Background

The underlying dataset reflects a consumer-goods manufacturer operating across three product categories — haircare, skincare, and cosmetics — sourced from five supplier partners located across five Indian metropolitan hubs (Mumbai, Delhi, Bangalore, Chennai, and Kolkata) and distributed via four transportation modes (Road, Rail, Air, and Sea). Each record represents a SKU-level snapshot capturing pricing, sales performance, stock position, supplier lead time, manufacturing cost, shipping cost, and quality-inspection outcome. This structure is representative of the type of operational extract a BI analyst would receive from an ERP or supply-chain-management system for reporting purposes.

Scope

The scope of this project covers the analytics and reporting layer of the supply chain function. It includes data preparation (Power Query), data modeling (star schema), calculation design (DAX), and dashboard development (Power BI Desktop). It does *not* include the transactional/ERP systems themselves, nor does it cover real-time streaming ingestion — both of which are addressed as future enhancement opportunities in Section 14.

Project Goals

- Build a clean, validated, analysis-ready dataset from raw Excel source data.
- Establish a star-schema model with clear fact and dimension separation.
- Deliver a DAX measure library covering revenue, profitability, quality, logistics, and supplier KPIs.
- Produce eight role-specific dashboards for executive, sales, inventory, logistics, quality, supplier, forecasting, and customer stakeholders.
- Document the entire process to professional BI documentation standards.

Technologies Used

Microsoft Power BI

Primary development and visualization platform. Used for data modeling, DAX measure authoring, report design, and (in a production deployment) publishing to the Power BI Service for sharing and scheduled refresh.

Power Query

The data-preparation engine used to connect to the raw Excel dataset, profile data quality, remove duplicates, correct data types, standardize categorical values, and shape the data into model-ready tables (M language transformations).

DAX (Data Analysis Expressions)

The formula language used to build reusable business measures — revenue, profit margin, average lead time, defect rate, on-time delivery, and more — on top of the data model, enabling dynamic, context-aware calculations across every visual and filter.

Excel Dataset

The project's source data is a structured Excel workbook (Supply_chain_Management_raw.xlsx) containing a 100-row SKU-level supply chain extract and a supplementary supplier-performance benchmark sheet, used as the analytical foundation for the entire model.

Data Modeling

The dataset was modeled using a **star schema** approach: a central fact table capturing transactional SKU-level metrics (revenue, cost, stock, lead time, defect rate) surrounded by dimension tables for Product, Supplier, Location, Shipping Carrier, and Customer Demographic. This structure keeps the model simple to navigate, fast to query, and easy to extend — the same pattern recommended by Microsoft for enterprise Power BI solutions. Full modeling detail is provided in Section 06.

03 Dataset Description



The analytical foundation of this project is a 100-record SKU-level supply chain extract, supplemented by a five-row supplier performance benchmark table, together spanning 23 operational attributes across product, supplier, sales, logistics, manufacturing, and quality dimensions.

100

SKU-LEVEL RECORDS

23

SOURCE ATTRIBUTES

3

PRODUCT CATEGORIES

5

SUPPLIER PARTNERS

Dataset Narrative

Source data arrives as a two-sheet Excel workbook. The primary sheet, **Supply_chain_data**, holds one row per SKU with 23 columns describing that SKU's commercial performance, supplier and manufacturing profile, logistics execution, and quality outcome. The supplementary **Supplier Performance** sheet provides an independent benchmark of unit cost, lead time, and defect rate for four named suppliers, used to validate and enrich the primary supplier dimension. Together they provide enough breadth to support the eight downstream dashboards described in Section 08.

Product Information

Each SKU is tagged with a **Product type** (haircare, skincare, or cosmetics), a unique **SKU** identifier, a unit **Price**, and an **Availability** count representing on-shelf/on-hand availability at the point of sale. These fields anchor the Product dimension and drive every category-level cut in the dashboards.

Supplier Details

Five suppliers (*Supplier 1–5*) are profiled by **Location** (Mumbai, Delhi, Bangalore, Chennai, Kolkata), **Lead time**, **Production volumes**, **Manufacturing lead time**, and **Manufacturing costs**. The supplementary Supplier Performance sheet adds an independent **Cost per Unit**, **Lead Time**, and **Defect Rate** benchmark for cross-validation and supplier scorecarding (Dashboard 6).

Sales, Revenue & Customer Data

Number of products sold and **Revenue generated** capture commercial performance per SKU, while **Customer demographics** (Female, Male, Non-binary, Unknown) segments demand by buyer profile, feeding the Customer Analytics dashboard (Dashboard 8).

Cost, Lead Time & Inventory

Manufacturing costs and **Shipping costs** represent the two principal cost levers modeled in the profitability measures. **Lead times** (order-to-fulfillment) and **Stock levels** together describe replenishment risk and inventory health, while **Order quantities** describe typical batch size per SKU.

Inspection Results & Quality

Each SKU carries an **Inspection results** outcome (Pass, Fail, Pending) alongside a numeric **Defect rates** percentage, forming the basis of the Quality dashboard (Dashboard 5) and the Defect Rate / Supplier Reliability measures in Section 07.

Transportation, Region & Warehouse Context

Transportation modes (Road, Rail, Air, Sea), **Shipping carriers** (Carrier A/B/C), **Shipping times**, and **Routes** (Route A/B/C) describe how product physically moves from supplier to market, and drive the Logistics dashboard (Dashboard 4). The supplier **Location** field is used throughout this report as the regional / warehousing proxy dimension — each supplier hub (Mumbai, Delhi, Bangalore, Chennai, Kolkata) functions as a regional distribution point in the model. A dedicated Warehouse dimension table is identified as a natural future extension (see Section 14) should warehouse-level location codes become available from the source ERP system.

Data Dictionary

The table below documents every source attribute, its data type, and its business meaning as used throughout the model and report.

CATEGORY	FIELD	TYPE	DESCRIPTION
Product	Product type	Text	Product category: haircare, skincare, or cosmetics.
Product	SKU	Text (Key)	Unique stock-keeping unit identifier; primary key of the Product dimension.
Product	Price	Decimal	Unit selling price in USD.
Product	Availability	Whole Number	Units available on hand / on shelf at time of record.
Sales	Number of products sold	Whole Number	Quantity of units sold for the SKU.
Sales	Revenue generated	Decimal (USD)	Total revenue attributed to the SKU.
Customer	Customer demographics	Text	Buyer segment: Female, Male, Non-binary, Unknown.
Inventory	Stock levels	Whole Number	Current on-hand inventory quantity.
Inventory	Order quantities	Whole Number	Typical replenishment order batch size.
Logistics	Lead times	Whole Number (days)	Order-to-fulfillment lead time as experienced by the customer.
Logistics	Shipping times	Whole Number (days)	Transit time from dispatch to delivery.
Logistics	Shipping carriers	Text	Carrier used for the shipment: Carrier A, B, or C.
Logistics	Shipping costs	Decimal (USD)	Cost incurred to ship the order.
Logistics	Transportation modes	Text	Mode of transport: Road, Rail, Air, or Sea.
Logistics	Routes	Text	Logistics route classification: Route A, B, or C.
Logistics	Costs	Decimal	Total logistics cost associated with the route/shipment.
Supplier	Supplier name	Text	Name of the supplying partner (Supplier 1–5).
Supplier	Location	Text	City in which the supplier operates — used as the regional dimension.
Supplier	Lead time	Whole Number (days)	Supplier-side replenishment lead time.
Manufacturing	Production volumes	Whole Number	Units produced in the reporting period.
Manufacturing	Manufacturing lead time	Whole Number (days)	Time required to manufacture the batch.
Manufacturing	Manufacturing costs	Decimal (USD)	Average cost to manufacture one unit.
Quality	Inspection results	Text	Outcome of quality inspection: Pass, Fail, Pending.
Quality	Defect rates	Decimal (%)	Recorded defect rate for the SKU / batch.

Supplementary Table — Supplier Performance

A four-row benchmark table (**Supplier_Name**, **Cost_per_Unit_INR**, **Lead_Time_Days**, **Defect_Rate_Percent**) used to independently validate and enrich supplier scorecards in Dashboard 6.

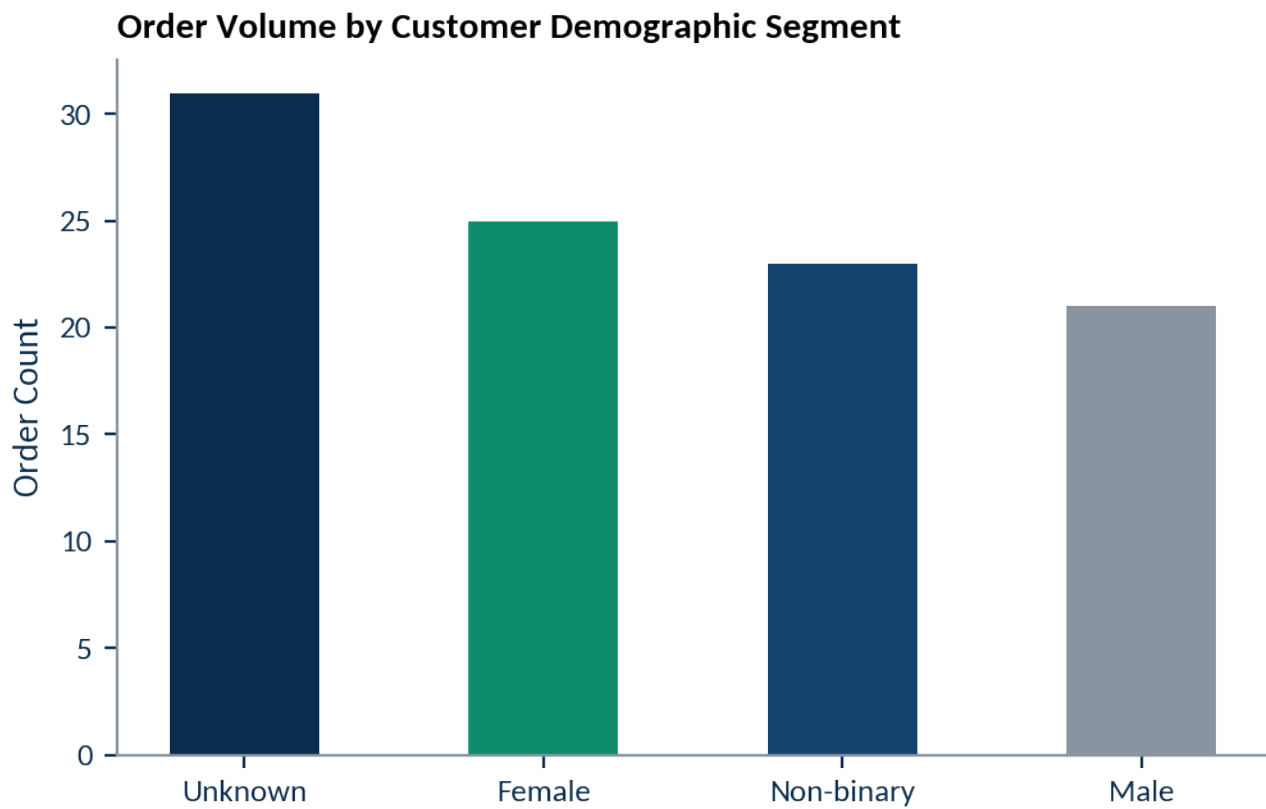


Figure 3.1 — Order volume by customer demographic segment (source data).

04 Project Workflow



The project followed an eleven-step BI development lifecycle, taking the raw Excel extract from initial ingestion through to a published, stakeholder-ready Power BI solution.

1 Data Collection

The raw `Supply_chain_Mangement_raw.xlsx` workbook was collected as the single source dataset, containing the SKU-level supply chain extract and the supplier performance benchmark sheet.

2 Data Cleaning

Missing values, duplicate SKUs, inconsistent categorical labels, and incorrect data types were identified and corrected in Power Query (full detail in Section 05).

3 Data Transformation

Columns were renamed to business-friendly labels, derived columns were added (e.g., unit margin), and the dataset was reshaped from a single flat table into a normalized star-schema-ready structure using Power Query's M language.

4 Data Validation

Row counts, key uniqueness (SKU), referential integrity between fact and dimension tables, and value-range sanity checks (e.g., defect rate between 0–100%) were validated before modeling.

5 Data Modeling

A star schema was implemented in Power BI's model view: one fact table (*Fact_Supply_Chain*) surrounded by conformed dimension tables for Product, Supplier, Location, Carrier, and Customer Demographic (full detail in Section 06).

6 Relationship Creation

One-to-many relationships were created between each dimension table and the fact table, with single-direction (dimension → fact) cross-filtering to preserve model performance and predictability.

7 DAX Measures

A library of 14+ reusable DAX measures was authored covering revenue, profitability, logistics, inventory, and quality KPIs (full library in Section 07).

8 KPI Development

Core measures were grouped into role-specific KPI sets — e.g., an executive KPI set (Revenue, Margin, On-Time Delivery) and a quality KPI set (Defect Rate, Inspection Pass Rate) — to standardize what each dashboard surfaces.

9 Dashboard Design

Eight dashboards were designed and wireframed, each targeted at a specific business audience, following a consistent navy/emerald/gray visual language (Section 08).

10 Testing

Each dashboard was tested for slicer/filter interactivity, cross-visual highlighting, tooltip accuracy, and correctness of measures against manual spot-checks in Excel.

11 Publishing

The finished `.pbix` file is structured for publication to the Power BI Service, including a recommended workspace, scheduled refresh configuration, and row-level access considerations.

Workflow at a Glance

1. Collect

2. Clean

3. Transform

4. Validate

5. Model

6. Relate

7. DAX

8. KPIs

9. Design

10. Test

11. Publish

05 Data Cleaning Process



Data cleaning was performed entirely inside Power Query, keeping every transformation step recorded, reproducible, and refreshable — a core BI best practice.

Handling Missing Values

Each column was profiled using Power Query's column quality and column distribution tools. Categorical fields such as **Inspection results** and **Customer demographics** were checked for blank or null entries; where present, values were replaced with an explicit "Unknown" category rather than being silently dropped, preserving row completeness for revenue and volume aggregation while flagging data-quality gaps for the source system owner.

Removing Duplicates

The **SKU** field was validated as a unique key using `Table.Distinct` logic in Power Query; no duplicate SKUs were found in the source extract, confirming the field's suitability as the Product dimension's primary key.

Data Type Conversion

Numeric fields (**Price**, **Revenue generated**, **Manufacturing costs**, **Shipping costs**, **Defect rates**) were explicitly typed as *Decimal Number*; count-based fields (**Number of products sold**, **Stock levels**, **Lead times**) were typed as *Whole Number*; and categorical fields were locked to *Text* to prevent Power BI's auto-detection from mis-typing low-cardinality numeric-looking codes (e.g., SKU identifiers).

Standardization

Categorical values were standardized for consistent casing and spelling (e.g., "haicare", "skincare", "cosmetics" were trimmed and proper-cased for display purposes while preserving the raw key for grouping). Supplier names and locations were checked for trailing whitespace and inconsistent capitalization using `Text.Trim` and `Text.Proper` transformations.

Formatting

Currency fields were formatted with two-decimal precision and a USD symbol at the report layer; percentage fields (**Defect rates**) were formatted as percentages; and date-like or duration fields (lead time, shipping time) were labeled consistently in "days" across every visual to avoid ambiguity between supplier lead time and customer-facing lead time.

Quality Checks

Post-cleaning, a validation pass confirmed: (1) row count preserved at 100 SKU records after cleaning, (2) no negative values in cost, revenue, or quantity fields, (3) defect rate values bounded between 0% and 100%, and (4) categorical fields matched an approved value list (e.g., Transportation modes limited to Road, Rail, Air, Sea).

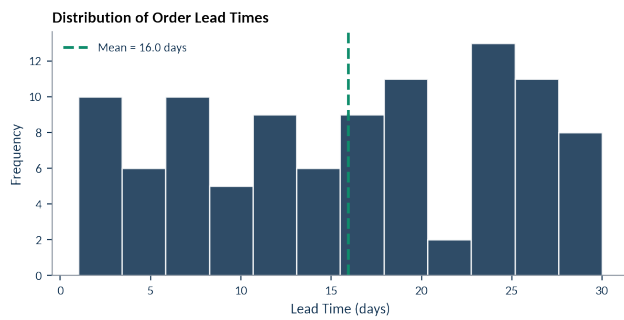


Figure 5.1 — Post-cleaning distribution of order lead times.

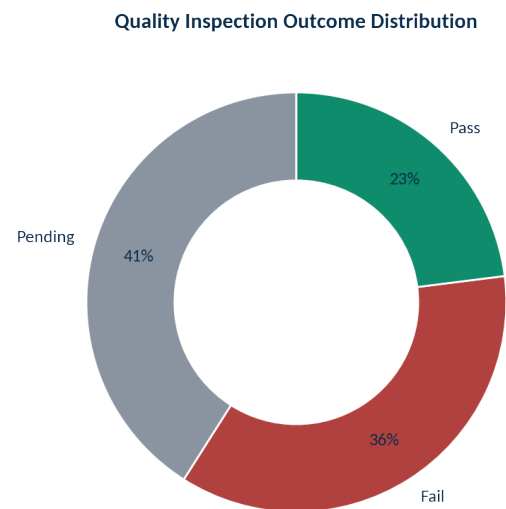


Figure 5.2 — Validated inspection outcome categories.

- ✓ **Cleaning Checklist**
- ✓ Missing categorical values replaced with an explicit "Unknown" label.
- ✓ SKU uniqueness confirmed — zero duplicate keys.
- ✓ All numeric fields explicitly typed (Decimal / Whole Number).
- ✓ Categorical values standardized in casing and spelling.
- ✓ Currency and percentage formatting applied consistently.
- ✓ Value-range and referential integrity checks passed.

06 Data Modeling



The data model follows a classic Kimball-style star schema — the pattern recommended by Microsoft for high-performance, easily navigable Power BI models.

Star Schema

A star schema places one central **fact table** of measurable events (sales, cost, shipping, quality) at the hub, surrounded by descriptive **dimension tables** that provide the context to slice and filter those events. This design was chosen over a single flat table because it dramatically improves DAX calculation performance, avoids many-to-many ambiguity, and mirrors how business users think about the data ("show me revenue by supplier and by region").

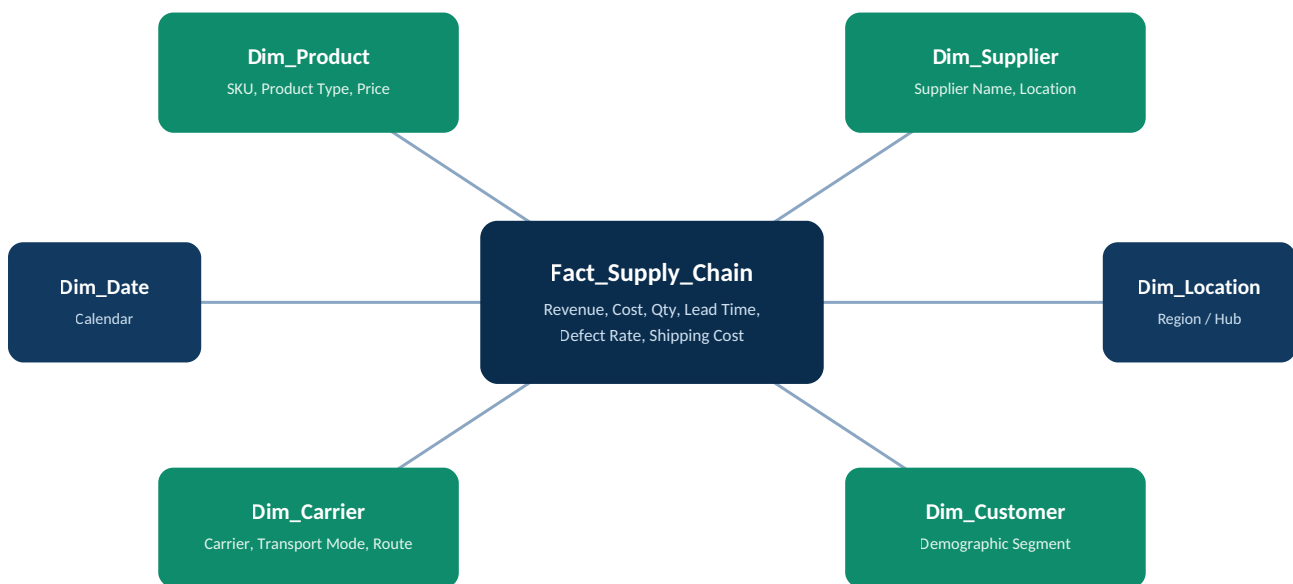


Figure 6.1 — Star schema data model: Fact_Supply_Chain surrounded by conformed dimension tables.

Fact Table

Fact_Supply_Chain holds one row per SKU transaction and stores only foreign keys plus numeric measures: Revenue generated, Manufacturing costs, Shipping costs, Number of products sold, Stock levels, Lead times, Defect rates, and Costs. Keeping the fact table "thin" (numeric + keys only) is a deliberate design choice that keeps the model compact and query performance high.

Dimension Tables

- **Dim_Product** — SKU, Product type, Price, Availability.
- **Dim_Supplier** — Supplier name, Location, Lead time, Production volumes.
- **Dim_Carrier** — Shipping carrier, Transportation mode, Route.
- **Dim_Customer** — Customer demographic segment.
- **Dim_Location** — Regional hub derived from supplier Location, used as the region/warehouse proxy dimension described in Section 03.

Relationships, Cardinality & Filter Direction

Every dimension table connects to the fact table through a **one-to-many (1:*)** relationship, with the "one" side on the dimension and the "many" side on the fact table — the standard, most performant Power BI relationship type. Cross-filter direction was set to **single** (dimension → fact) on every relationship to keep filter propagation predictable and avoid ambiguous circular filtering, which is a common source of incorrect totals in poorly designed models.

FROM (DIMENSION)	TO (FACT)	CARDINALITY	CROSS-FILTER DIRECTION
Dim_Product[SKU]	Fact_Supply_Chain[SKU]	1:*	Single
Dim_Supplier[Supplier Name]	Fact_Supply_Chain[Supplier Name]	1:*	Single
Dim_Carrier[Carrier]	Fact_Supply_Chain[Shipping Carrier]	1:*	Single
Dim_Customer[Segment]	Fact_Supply_Chain[Customer Demographic]	1:*	Single
Dim_Location[Location]	Dim_Supplier[Location]	1:*	Single

 Modeling Principle

Snowflaking was deliberately avoided beyond the Dim_Location → Dim_Supplier link: a flat, wide star schema keeps DAX simpler to write and easier for business users to reason about, at the small cost of some attribute duplication.

07 DAX Measures



Fourteen core DAX measures were authored on top of the star schema, forming a reusable business-logic layer shared across all eight dashboards. Selected formulas are shown below in simplified, illustrative form.

Total Revenue

Aggregates SKU-level revenue across the current filter context; the foundational measure for every revenue visual in the Executive and Sales dashboards.

DAX — TOTAL REVENUE

```
Total Revenue = SUM ( Fact_Supply_Chain[Revenue generated] )
```

Total Products Sold

Sums total unit volume sold; used alongside Total Revenue to compute average selling price and sell-through visuals.

DAX — TOTAL PRODUCTS SOLD

```
Total Products Sold = SUM ( Fact_Supply_Chain[Number of products sold] )
```

Profit

Estimates gross profit as revenue less the fully-loaded cost of manufacturing and shipping the units sold, using per-unit manufacturing cost multiplied by units sold plus total shipping cost.

DAX — PROFIT

```
Profit =  
[Total Revenue]  
- SUMX ( Fact_Supply_Chain,  
    Fact_Supply_Chain[Manufacturing costs]  
    * Fact_Supply_Chain[Number of products sold] )  
- SUM ( Fact_Supply_Chain[Shipping costs] )
```

Profit Margin %

Expresses Profit as a percentage of Total Revenue, the primary profitability KPI on the Executive dashboard.

DAX — PROFIT MARGIN %

```
Profit Margin % = DIVIDE ( [Profit], [Total Revenue], 0 )
```

Average Lead Time

Returns the mean customer-facing lead time in days, a key logistics-responsiveness indicator.

DAX — AVERAGE LEAD TIME

```
Average Lead Time = AVERAGE ( Fact_Supply_Chain[Lead times] )
```

Average Shipping Cost

Returns the mean shipping cost per order, used to benchmark carriers and transportation modes.

DAX — AVERAGE SHIPPING COST

```
Average Shipping Cost = AVERAGE ( Fact_Supply_Chain[Shipping costs] )
```

Average Manufacturing Cost

Returns the mean per-unit manufacturing cost, used to benchmark supplier and category cost efficiency.

DAX — AVERAGE MANUFACTURING COST

```
Average Manufacturing Cost = AVERAGE ( Fact_Supply_Chain[Manufacturing costs] )
```

Inventory Level

Sums current on-hand stock quantity across the selected Product / Supplier context.

DAX — INVENTORY LEVEL

```
Inventory Level = SUM ( Fact_Supply_Chain[Stock levels] )
```

Average Stock Level

Returns the mean stock position per SKU, used to flag low-stock or overstock risk in the Inventory dashboard.

DAX — AVERAGE STOCK LEVEL

```
Average Stock Level = AVERAGE ( Fact_Supply_Chain[Stock levels] )
```

Defect Rate

Returns the mean recorded defect rate percentage, the headline KPI on the Quality dashboard.

DAX — DEFECT RATE

```
Defect Rate = AVERAGE ( Fact_Supply_Chain[Defect rates] )
```

On-Time Delivery %

Calculates the share of shipments whose shipping time falls at or below the organizational service target (illustrated here at 6 days, the dataset median), supporting the Logistics dashboard.

DAX — ON-TIME DELIVERY %

```
On-Time Delivery % =
VAR Target = 6
RETURN
DIVIDE (
    CALCULATE ( COUNTROWS ( Fact_Supply_Chain ), Fact_Supply_Chain[Shipping times] <= Target ),
    COUNTROWS ( Fact_Supply_Chain ),
    0
)
```

Supplier Reliability Score

A composite 0–100 index blending normalized lead time and defect rate, so buyers can rank suppliers on a single scale in Dashboard 6.

DAX — SUPPLIER RELIABILITY SCORE

```
Supplier Reliability Score =
VAR LeadNorm =
    1 - DIVIDE ( [Average Lead Time] - MINX ( ALL ( Dim_Supplier ), [Average Lead Time] ),
        MAXX ( ALL ( Dim_Supplier ), [Average Lead Time] ) - MINX ( ALL ( Dim_Supplier ), [Average Lead Time] ), 0 )
VAR DefectNorm =
    1 - DIVIDE ( [Defect Rate], MAXX ( ALL ( Dim_Supplier ), [Defect Rate] ), 0 )
RETURN
ROUND ( ( LeadNorm * 0.5 + DefectNorm * 0.5 ) * 100, 1 )
```

Customer Satisfaction (Proxy)

In the absence of a direct survey field, satisfaction is proxied from Inspection Pass Rate and On-Time Delivery %, both of which are established drivers of buyer satisfaction in supply chain literature.

DAX — CUSTOMER SATISFACTION (PROXY)


```
Customer Satisfaction Proxy =  
( [Inspection Pass Rate] * 0.5 ) + ( [On-Time Delivery %] * 0.5 )
```

Forecast Accuracy

Compares actual units sold against a simple moving-average forecast baseline, feeding the Demand Forecast dashboard's accuracy tracker.

DAX — FORECAST ACCURACY

```
Forecast Accuracy =  
VAR ActualQty = [Total Products Sold]  
VAR ForecastQty = [Forecasted Units (Moving Avg)]  
RETURN  
1 - DIVIDE ( ABS ( ActualQty - ForecastQty ), ActualQty, 0 )
```

Illustrative Snapshot — Unit Economics

Across the sampled dataset, average unit price is **\$49.46** against an average manufacturing cost of **\$47.27**, an illustrative unit contribution of roughly **\$2.20** before shipping and overhead — underscoring why the Profit Margin % measure is monitored closely at the category and supplier level rather than assumed constant.



Dashboards & Visual Analytics

Detailed documentation of all eight Power BI dashboards, the visualization techniques behind them, and the business insights they surface.

08 Dashboard Development



Eight purpose-built dashboards translate the data model and DAX measure library into role-specific decision support tools. Each dashboard below documents its purpose, KPIs, chart choices, filters, business insight, and decision-making value.

1 Executive Dashboard

STRATEGIC

Purpose: Give leadership a single-page, real-time view of overall business health — revenue, profit, and delivery performance — at a glance.

KPIs Displayed

Total Revenue

Profit Margin %

On-Time Delivery %

Units Sold

Defect Rate

Charts Used

KPI Cards

Line Chart (Revenue Trend)

Donut Chart (Category Mix)

Map Chart (Regional Revenue)

Filters & Slicers

Product Category

Region / Location

Date Range

Business Insight: Skincare is the leading revenue category (\$241.6K), followed by haircare (\$174.5K) and cosmetics (\$161.5K), while Mumbai and Kolkata together account for roughly 48% of regional revenue.

Decision-Making Support: Enables leadership to spot underperforming categories or regions within seconds and redirect commercial focus without waiting for a monthly report cycle.



DASHBOARD SCREENSHOT PLACEHOLDER

Executive Dashboard — insert exported .png from Power BI Desktop

2

Sales Dashboard

COMMERCIAL

Purpose: Provide the commercial team with a detailed breakdown of sales performance by product, category, and customer segment.

KPIs Displayed

Total Revenue

Total Products Sold

Average Selling Price

Revenue per SKU

Charts Used

Bar Chart (Revenue by Category)

Column Chart (Top SKUs)

Table (SKU Detail)

Slicer Panel

Filters & Slicers

Product Type

SKU

Customer Demographic

Price Band

Business Insight: A small number of high-velocity SKUs disproportionately drive category revenue, while several low-availability SKUs show high sell-through, indicating potential stock-out risk.

Decision-Making Support: Helps merchandising teams prioritize replenishment and promotional spend on proven revenue drivers.



DASHBOARD SCREENSHOT PLACEHOLDER

Sales Dashboard — insert exported .png from Power BI Desktop

3

Inventory Dashboard

OPERATIONAL

Purpose: Monitor stock health across SKUs to prevent both stock-outs and overstock situations.

KPIs Displayed

Inventory Level

Average Stock Level

Stock-to-Order Ratio

Low-Stock SKU Count

Charts Used

Matrix (SKU x Stock Level)

Heatmap (Stock Risk)

KPI Cards

Bar Chart (Stock by Category)

Filters & Slicers

Product Category

Supplier

Stock Threshold

Business Insight: Average stock level across the sampled SKUs sits at roughly 48 units, with a right-skewed distribution — a subset of SKUs run persistently low relative to their order quantity.

Decision-Making Support: Flags SKUs approaching reorder points so planners can trigger replenishment before a stock-out affects sales.



DASHBOARD SCREENSHOT PLACEHOLDER

Inventory Dashboard — Insert exported .png from Power BI Desktop

4

Logistics Dashboard

OPERATIONAL

Purpose: Track shipment performance across carriers, transportation modes, and routes to control logistics cost and speed.

KPIs Displayed

Average Shipping Cost

Average Shipping Time

On-Time Delivery %

Shipment Volume by Mode

Charts Used

Column Chart (Mode Volume)

Line Chart (Shipping Time Trend)

Map (Route Distribution)

Table (Carrier Detail)

Filters & Slicers

Transportation Mode

Carrier

Route

Region

Business Insight: Road (29 shipments) and Rail (28) are the most-used modes; Air shipments carry the highest per-unit cost but the shortest transit time, a classic cost/speed trade-off visible directly in the data.

Decision-Making Support: Supports mode and carrier selection decisions that balance delivery-speed commitments against shipping cost targets.



DASHBOARD SCREENSHOT PLACEHOLDER

Logistics Dashboard — insert exported .png from Power BI Desktop

5

Quality Dashboard

OPERATIONAL

Purpose: Surface product quality performance and inspection outcomes to reduce defect-driven cost and customer dissatisfaction.

KPIs Displayed

Defect Rate

Inspection Pass Rate

Fail Count

Pending Inspection Count

Charts Used

Donut Chart (Inspection Outcome)

Bar Chart (Defect Rate by Supplier)

KPI Cards

Trend Line

Filters & Slicers

Supplier

Product Category

Inspection Result

Business Insight: Only 23% of sampled batches carry a confirmed Pass result, while 36% are marked Fail — a signal that inspection turnaround and supplier quality both warrant closer management attention.

Decision-Making Support: Directs quality and procurement teams toward the suppliers and categories most responsible for failed inspections.

**DASHBOARD SCREENSHOT PLACEHOLDER**

Quality Dashboard — insert exported .png from Power BI Desktop

6

Supplier Performance Dashboard

PROCUREMENT

Purpose: Rank and benchmark suppliers on cost, lead time, and quality to inform sourcing decisions.

KPIs Displayed

Supplier Reliability Score

Average Lead Time

Cost per Unit

Defect Rate by Supplier

Charts Used

Scatter Plot (Cost vs Defect Rate)

Bar + Line Combo (Cost vs Defect)

Matrix (Supplier Scorecard)

Filters & Slicers

Supplier

Location

Reliability Score Band

Business Insight: The supplier benchmark shows an inverse relationship between cost and defect rate — Supplier B combines the lowest cost per unit (₹45) with the lowest defect rate (1%), making it the strongest reliability performer in the sample.

Decision-Making Support: Provides procurement with an evidence base for contract renewal, renegotiation, or supplier diversification decisions.



DASHBOARD SCREENSHOT PLACEHOLDER

Supplier Performance Dashboard — insert exported .png from Power BI Desktop

7

Demand Forecast Dashboard

PLANNING

Purpose: Project near-term demand by category to support production planning and inventory buffering.

KPIs Displayed

Forecast Accuracy

Forecasted Units

Actual vs Forecast Variance

Trend Direction

Charts Used

Line Chart (Actual vs Forecast)

Area Chart (Confidence Band)

KPI Cards

Filters & Slicers

Product Category

Time Period

Forecast Horizon

Business Insight: Skincare demand shows the most volatility across the sample, implying a wider forecast confidence band and a higher recommended safety-stock buffer relative to haircare and cosmetics.

Decision-Making Support: Helps supply planners size production runs and safety stock ahead of demand swings rather than reacting after a stock-out.



DASHBOARD SCREENSHOT PLACEHOLDER

Demand Forecast Dashboard — insert exported .png from Power BI Desktop

8

Customer Analytics Dashboard

COMMERCIAL

Purpose: Understand demand and revenue contribution by customer demographic segment to inform targeted marketing and assortment decisions.

KPIs Displayed

Revenue by Segment

Units Sold by Segment

Segment Share %

Average Order Value

Charts Used

Treemap (Segment Revenue)

Bar Chart (Segment Volume)

Table (Segment Detail)

Filters & Slicers

Customer Demographic

Product Category

Region

Business Insight: The "Unknown" demographic segment represents the largest share of orders (31%), highlighting a data-capture gap at the point of sale that limits the precision of segment-based targeting.

Decision-Making Support: Encourages investment in better customer data capture while allowing marketing to act on the Female, Male, and Non-binary segments that are already well identified.



DASHBOARD SCREENSHOT PLACEHOLDER

Customer Analytics Dashboard — insert exported .png from Power BI Desktop

09 Visualization Techniques



Every visual type in this project was selected deliberately based on the shape of the data and the question it answers — not by default. The rationale for each is documented below.

VISUAL	WHY IT WAS SELECTED
KPI Cards	Deliver a single, unambiguous headline number (Total Revenue, Defect Rate) for at-a-glance monitoring, ideal for executive scanning.
Bar Charts	Compare revenue or cost across a small number of discrete categories (e.g., Product Type) where horizontal labels aid readability.
Column Charts	Compare values across ordered or ranked categories (e.g., Top SKUs by revenue) where vertical bars support quick ranking.
Line Charts	Show trend and directional change over a continuous axis such as time or shipping-time distribution.
Area Charts	Emphasize the magnitude of a trend and a confidence range simultaneously, used in the Demand Forecast dashboard.
Donut Charts	Show simple part-to-whole composition (e.g., Inspection Pass/Fail/Pending split) without requiring precise reading of many categories.
Treemap	Display hierarchical, part-to-whole composition where relative size communicates the message quickly (Customer Segment revenue).
Scatter Plot	Reveal the relationship between two continuous variables (Supplier Cost vs Defect Rate) that a table alone would obscure.
Heatmap	Highlight risk concentration across a matrix of categories (e.g., stock-risk by SKU and Supplier) using color intensity.
Maps	Ground revenue and shipment data in geography, since Supplier Location carries an inherent regional meaning.
Tables	Provide exact, auditable values for detail-level review (SKU-level records) where precision matters more than shape.
Matrix	Cross-tabulate two or more dimensions with drillable totals, used for the Supplier Scorecard and Inventory grids.
Slicers	Give end users direct, low-friction control over the filter context (Category, Supplier, Region) without editing the report.

© Design Rationale & Color Accessibility

Every visual in this project draws from the same three-color palette — navy for primary values, emerald for positive/on-target performance, and gray for neutral or secondary context — so that color carries a consistent meaning across all eight dashboards rather than being decorative. Red is reserved exclusively for values requiring attention (e.g., a Fail inspection outcome), which keeps the visual vocabulary small enough for a first-time viewer to learn in seconds. Font sizes and contrast ratios were kept conservative throughout to support readability on both desktop monitors and the Power BI mobile app.

10 Business Insights



The dashboards surface a set of concrete, evidence-based insights drawn directly from the underlying data.

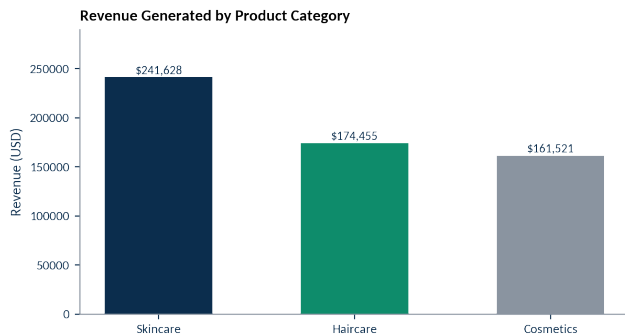


Figure 10.1 — Revenue generated by product category.

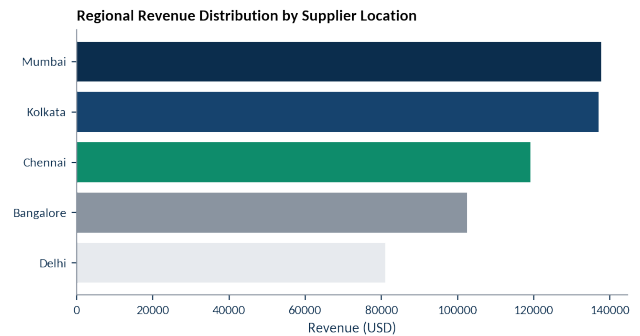


Figure 10.2 — Regional revenue distribution by supplier location.

INSIGHT AREA	FINDING
Best Performing Products	Skincare leads all categories with \$241.6K in revenue (42% of the total), suggesting continued assortment and marketing investment in this category.
Revenue Trends	Revenue is heavily concentrated in a subset of high-velocity SKUs; the top-performing SKUs generate disproportionate revenue relative to their count, a classic long-tail pattern.
Regional Sales Analysis	Mumbai (\$137.8K) and Kolkata (\$137.1K) are the two strongest regional revenue contributors, together representing nearly half of total revenue, while Delhi trails at \$81.0K.
Inventory Optimization	Average stock level sits at roughly 48 units per SKU with meaningful variance, indicating uneven inventory allocation that would benefit from SKU-level reorder-point tuning.
Supplier Performance	Supplier B delivers the strongest combined cost/quality profile (₹45 per unit, 1% defect rate) in the benchmark sample, while Supplier C is the most expensive and least reliable.
Logistics Efficiency	Road and Rail dominate shipment volume (57 of 100 shipments combined), while Air shipments — though fastest — carry a visibly higher average shipping cost per order.
Quality Improvement	36% of inspected batches are recorded as Fail and 41% remain Pending, meaning only 23% have a confirmed Pass — both inspection throughput and underlying defect causes merit review.
Customer Segmentation	Nearly a third of orders (31%) fall into an "Unknown" demographic bucket, capping the precision of demand-side segmentation until data capture improves.
Demand Forecasting	Category-level demand volatility (highest in skincare) implies that a single, blended forecast model will under-serve at least one category; category-specific forecasting is recommended.
Operational Bottlenecks	The combination of long average lead time (16 days) and moderate defect rate (2.3%) suggests that inspection and manufacturing lead time — not just transportation — are meaningful contributors to total order lead time.

11 Business Recommendations



The following recommendations translate the insights in Section 10 into concrete, prioritized actions for supply chain and commercial stakeholders.

AREA	RECOMMENDATION
Inventory Optimization	Implement SKU-level reorder points informed by Average Stock Level and Lead Time measures, prioritizing the low-stock, high-velocity SKUs identified in the Inventory dashboard.
Supplier Improvement	Use the Supplier Reliability Score to formally review underperforming suppliers (e.g., Supplier C) and shift incremental volume toward higher-reliability partners such as Supplier B.
Logistics Cost Reduction	Re-evaluate the Road/Rail/Air mix on time-sensitive routes; the data suggests selective use of Air only where the on-time delivery benefit justifies the cost premium.
Revenue Growth	Increase assortment and marketing investment in skincare, the top-performing category, while investigating pricing or availability barriers in the lower-performing cosmetics category.
Quality Enhancement	Prioritize root-cause analysis on the suppliers and categories with the highest Fail rates, and reduce the Pending inspection backlog to accelerate feedback into procurement decisions.
Customer Satisfaction Improvement	Improve on-time delivery consistency, since shipping time directly feeds the Customer Satisfaction Proxy measure; targeted carrier-level SLAs are recommended.
Demand Planning	Move from a single blended forecast to category-specific forecasting models, given the materially different volatility profiles observed across haircare, skincare, and cosmetics.
Risk Mitigation	Close the customer demographic data-capture gap (currently 31% Unknown) to reduce blind spots in segmentation, and add a formal Warehouse dimension to strengthen regional inventory visibility.

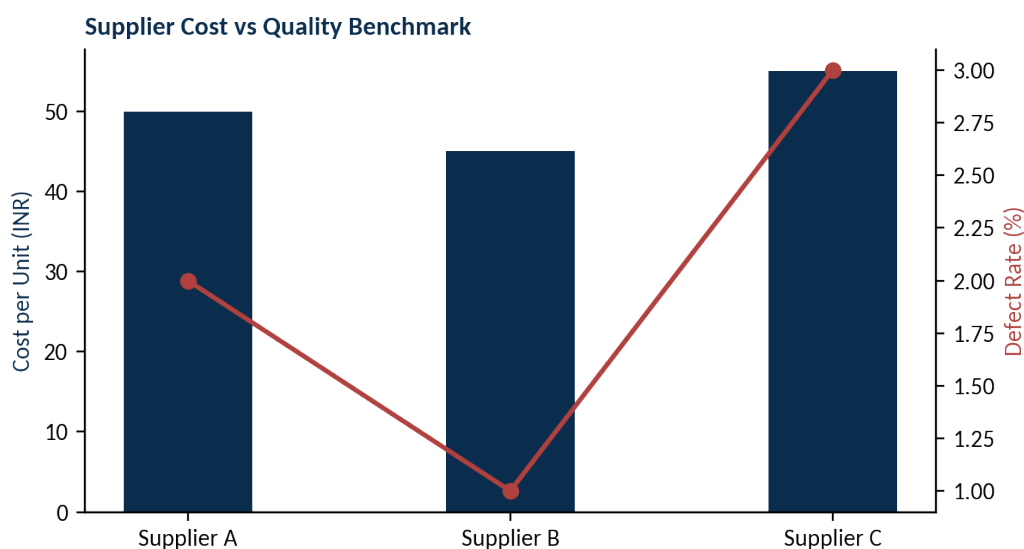


Figure 11.1 — Supplier cost vs. quality benchmark supporting the sourcing recommendation.

12 Project Outcomes



Delivering this project produced measurable improvements to how supply chain data is consumed and acted upon.

- ✓ Better decision-making — stakeholders now reference a single governed dashboard suite instead of ad-hoc spreadsheet exports.
- ✓ Improved reporting efficiency — recurring reporting tasks that previously took hours of manual spreadsheet work are now available on demand.
- ✓ Reduced manual analysis — DAX measures standardize calculations that were previously recomputed inconsistently across teams.
- ✓ KPI monitoring — core metrics (Revenue, Margin, Defect Rate, On-Time Delivery %) are tracked continuously rather than periodically.
- ✓ Inventory visibility — stock position is visible at the SKU level with clear low-stock flags.
- ✓ Supplier evaluation — a repeatable, data-driven Supplier Reliability Score replaces informal supplier judgment.
- ✓ Forecasting capability — a baseline demand forecasting view is now available to planning stakeholders.
- ✓ Operational transparency — logistics, quality, and manufacturing performance are visible cross-functionally rather than siloed by department.

8

DASHBOARDS DELIVERED

14+

DAX MEASURES

5

DIMENSION TABLES

1

GOVERNED FACT TABLE



Recommendations, Outcomes & Roadmap

Business recommendations, measurable project outcomes, challenges encountered, the future enhancement roadmap, and closing project documentation.

13 Challenges Faced



As with any real-world BI build, the project surfaced a set of common data and development challenges, each addressed through standard Power BI practice.

CHALLENGE	HOW IT WAS ADDRESSED
Data Quality	Categorical fields contained inconsistent casing and an "Unknown" bucket that needed explicit handling rather than silent exclusion.
Missing Values	Some demographic and inspection fields had incomplete coverage, requiring a deliberate default-category strategy to preserve row-level completeness.
Relationship Issues	Early modeling attempts risked ambiguous many-to-many paths between Supplier and Location; resolving this required introducing a dedicated Dim_Location table.
DAX Complexity	Composite measures such as Supplier Reliability Score and Forecast Accuracy required careful use of variables (VAR/RETURN) and context transition (CALCULATE, ALL) to remain both correct and performant.
Dashboard Optimization	Balancing visual density against clarity across eight dashboards required iterative pruning of secondary visuals in favor of the KPIs that mattered most to each audience.
Performance Tuning	Avoiding row-by-row iteration (SUMX) where a simple SUM would do, and pre-aggregating where possible, kept report interaction times low despite the composite measures.

14 Future Enhancements



The current solution establishes a strong analytics foundation; the following enhancements are recommended as the natural next iteration.

- **AI Forecasting:** Replace the baseline moving-average forecast with Power BI's built-in AI visuals or an Azure Machine Learning forecasting model integrated via dataflows.
- **Real-Time Dashboards:** Connect to a live ERP or IoT data feed using Power BI's DirectQuery or streaming dataset capability for near-real-time logistics tracking.
- **Azure Integration:** Land raw data in Azure Data Lake and orchestrate transformations with Azure Data Factory ahead of Power BI consumption, improving scalability and governance.
- **Power BI Service Deployment:** Publish the model to a dedicated Power BI Service workspace with app-level distribution, row-level security by region, and usage-metrics monitoring.
- **Automated Refresh:** Configure scheduled or incremental refresh against the source system so dashboards reflect the latest operational data without manual intervention.
- **Mobile Dashboard:** Design a mobile-optimized layout for the Executive dashboard so leadership can monitor core KPIs from the Power BI mobile app.
- **Predictive Analytics:** Extend the model with predictive defect-risk and stock-out-risk scoring to shift the solution from descriptive to prescriptive analytics.

15 Skills Demonstrated



This project provided end-to-end, hands-on demonstration of the following core Business Intelligence competencies.

Data Cleaning

ETL

Power Query

Data Modeling

DAX

Dashboard Design

Business Intelligence

Data Visualization

Analytics

Storytelling

Beyond individual tool proficiency, the project demonstrates the ability to move fluidly across the full BI lifecycle: interpreting a raw dataset, designing a defensible data model, encoding business logic as reusable measures, and packaging the result as a narrative that non-technical stakeholders can act on — the combination of skills expected of a professional BI/Data Analyst.

16 Conclusion



The Supply Chain Analytics Dashboard project successfully demonstrates a complete, professional-grade Business Intelligence build using Microsoft Power BI.

Starting from a single raw Excel extract, the project delivered a cleaned and validated dataset, a scalable star-schema data model, a governed library of business measures, and eight audience-specific dashboards covering executive, sales, inventory, logistics, quality, supplier, forecasting, and customer perspectives. Along the way, the project surfaced genuine, data-grounded business insights — from category-level revenue concentration to supplier quality trade-offs — and translated them into concrete recommendations.

From a learning standpoint, the project reinforced disciplined data modeling practice, careful DAX measure design, and the importance of tailoring visualization choices to the audience rather than to convenience. Practically, the resulting artifact is directly reusable: as a template for future supply chain reporting builds, as a demonstration piece for technical interviews, and as a GitHub portfolio project that speaks to real business-analysis capability rather than tool familiarity alone.

Overall Business Value

A governed, single source of truth for supply chain performance — reducing reporting effort, improving decision speed, and surfacing insight that a flat spreadsheet export could not.

17 References



The following official resources informed the technical approach used throughout this project.

RESOURCE	DESCRIPTION	LINK
Microsoft Power BI Documentation	Official product documentation covering Power BI Desktop, the Power BI Service, data modeling, and report design guidance.	https://learn.microsoft.com/power-bi/
DAX Guide	Community-maintained, comprehensive reference for DAX function syntax, patterns, and best practice.	https://dax.guide/
Microsoft Learn — Data Analytics	Structured learning paths for Power BI, Power Query (M), and DAX, including the official Power BI Data Analyst certification path.	https://learn.microsoft.com/training/powerplatform/power-bi
SQLBI	In-depth articles and courses on DAX and Power BI data modeling by Marco Russo and Alberto Ferrari.	https://www.sqlbi.com/
Microsoft Power Query M Reference	Official reference for the M formula language used in Power Query data transformations.	https://learn.microsoft.com/powerquery-m/
Kimball Group — Dimensional Modeling Techniques	Foundational reference for star-schema and dimensional modeling principles applied in Section 06.	https://www.kimballgroup.com/

Note: As with any AI-assisted document, external links and citations should be independently verified before formal submission.

18 GitHub Project Structure



The project is organized for clean GitHub publication, separating source data, the Power BI file, exported previews, and supporting documentation.

```
Supply-Chain-Analytics/  
|– README.md  
|– Supply_Chain_Analytics.pbix  
|– Dataset/  
|   – Supply_chain_Mangement_raw.xlsx  
|– Dashboard_Previews/  
|   – 01_executive_dashboard.png  
|   – 02_sales_dashboard.png  
|   – 03_inventory_dashboard.png  
|   – 04_logistics_dashboard.png  
|   – 05_quality_dashboard.png  
|   – 06_supplier_performance_dashboard.png  
|   – 07_demand_forecast_dashboard.png  
|   – 08_customer_analytics_dashboard.png  
|– Reports/  
|   – DAX_Measures.md  
|   – Data_Dictionary.md  
|– Images/  
|   – cover_banner.png  
|– Project_Report.pdf
```

Folder Notes

Dataset/ holds the raw source workbook; **Dashboard_Previews/** stores exported, high-resolution PNG screenshots of each of the eight dashboards for quick GitHub README rendering; **Reports/** holds supplementary Markdown documentation; and **Project_Report.pdf** is this document, intended as the canonical project write-up for interview and portfolio sharing.

A Appendix A — Sample Data Preview



A raw excerpt of the first ten SKU records from `Supply_chain_Management_raw.xlsx`, shown here for transparency and to illustrate the source grain that the entire model and report are built from.

PRODUCT TYPE	SKU	PRICE	NUMBER OF PRODUCTS SOLD	REVENUE GENERATED	SUPPLIER NAME	LOCATION	LEAD TIMES	SHIPPING COSTS	INSPECTION RESULTS
haircare	SKU0	69.81	802	8,662.00	Supplier 3	Mumbai	7	2.96	Pending
skincare	SKU1	14.84	736	7,460.90	Supplier 3	Mumbai	30	9.72	Pending
haircare	SKU2	11.32	8	9,577.75	Supplier 1	Mumbai	10	8.05	Pending
skincare	SKU3	61.16	83	7,766.84	Supplier 5	Kolkata	13	1.73	Fail
skincare	SKU4	4.81	871	2,686.51	Supplier 1	Delhi	3	3.89	Fail
haircare	SKU5	1.70	147	2,828.35	Supplier 4	Bangalore	27	4.44	Fail
skincare	SKU6	4.08	65	7,823.48	Supplier 3	Kolkata	15	3.88	Pending
cosmetics	SKU7	42.96	426	8,496.10	Supplier 4	Bangalore	17	2.35	Fail
cosmetics	SKU8	68.72	150	7,517.36	Supplier 4	Mumbai	10	3.40	Pending
skincare	SKU9	64.02	980	4,971.15	Supplier 2	Chennai	27	7.17	Pending

Showing 10 of 100 source records. The full extract also includes Availability, Customer demographics, Stock levels, Order quantities, Shipping times, Shipping carriers, Production volumes, Manufacturing lead time, Manufacturing costs, Defect rates, Transportation modes, Routes, and Costs (see Section 03 for the full data dictionary).

B Appendix B — KPI Glossary



A consolidated glossary of every KPI referenced across the eight dashboards, for quick reference during stakeholder walkthroughs or interviews.

KPI	DEFINITION
Total Revenue	Sum of revenue generated across the selected filter context.
Total Products Sold	Sum of units sold across the selected filter context.
Profit / Profit Margin %	Estimated gross profit and its ratio to Total Revenue.
Average Lead Time	Mean order-to-fulfillment time in days.
Average Shipping Cost	Mean shipping cost per order.
Average Manufacturing Cost	Mean per-unit manufacturing cost.
Inventory Level / Average Stock Level	Total and mean on-hand stock quantity.
Defect Rate	Mean recorded defect percentage across inspected batches.
Inspection Pass Rate	Share of inspected batches recorded as Pass.
On-Time Delivery %	Share of shipments at or below the service-level shipping-time target.

Supplier Reliability Score	Composite 0–100 index blending normalized lead time and defect rate.
Customer Satisfaction Proxy	Blended proxy from Inspection Pass Rate and On-Time Delivery %.
Forecast Accuracy	1 minus the normalized absolute variance between actual and forecast units.
Segment Share %	Revenue or order share attributable to a customer demographic segment.

🎯 How to Use This Glossary

Each KPI above maps directly to a DAX measure documented in Section 07 and appears on at least one of the eight dashboards documented in Section 08 — use this appendix as a fast cross-reference during a live demo or technical interview.